Stat Analysis & Algebraic Reasoning

Unit 2: Describing One Variable

Practice Test — Study Copy— Answer Key

Keep in Mind

This is a practice test. It mirrors the real Unit 2 test in length, format, and the ideas it covers, but the numbers and contexts differ. Work every problem with no notes, then check your answers against the key. **25 questions, 100 points:** Part A 10, Part B 12, Part C 48, Part D 30.

Part A — Vocabulary — 10 questions, 1 point each

Set 1 — Displays and Shape. Write the letter of the matching description next to each term.

1. **C** Distribution
2. **A** Stemplot
3. **E** Histogram
4. **B** Skewed right
5. **D** Skewed left

A. Each value split into a stem and a leaf, read with a key.

B. The long tail stretches toward the larger values.

C. What values a variable takes and how often it takes each one.

D. The long tail stretches toward the smaller values.

E. Bars over equal-width bins; each bar's height is a count.

Set 2 — Measures and Outliers. Same directions.

6. **D** Median
7. **E** Interquartile range
8. **B** Standard deviation
9. **C** Outlier
10. **A** Resistant

A. Barely moved by one extreme value.

B. About the typical distance of the values from their mean.

C. A value below $Q_1 - 1.5 \times \text{IQR}$ or above $Q_3 + 1.5 \times \text{IQR}$.

D. The middle value of the ordered data.

E. $Q_3 - Q_1$: the spread of the middle half of the data.

Part B — Multiple Choice — 6 questions, 2 points each

11. A histogram of the minutes 60 Granite Falls students spent at a lunchtime club has its tallest bars between 0 and 15 minutes, then shorter and shorter bars out to 60 minutes. The distribution is

(A) skewed left, because the tallest bars are on the left.

(B) skewed right, because the long tail stretches toward the larger values.

(C) symmetric, because it has only one peak.

(D) skewed left, because most students spent only a little time.

Letter: **B**

12. House prices in a town are strongly skewed right by a few very expensive homes. The best measure of a typical price is

(A) the median, because the few expensive homes barely move it.

(B) the mean, because it uses every price.

(C) the range, because it shows the most expensive home.

(D) the standard deviation, because it measures typical distance.

Letter: **A**

13. Class P's quiz scores run from 88 to 96 with $s = 2.5$. Class Q's run from 60 to 95 with $s = 10$. Which statement is true?

(A) Class P should have the larger s, because its scores are higher.

(B) The two standard deviations cannot be compared.

(C) Class Q's scores typically sit farther from their own mean than Class P's.

(D) Class Q scored higher, because its standard deviation is larger.

Letter: **C**

14. Every value in a data set is increased by 5. What happens to the mean and the standard deviation?

(A) Both increase by 5.

(B) The mean increases by 5; the standard deviation does not change.

(C) The mean stays the same; the standard deviation increases by 5.

(D) Neither one changes.

Letter: **B**

15. One very large outlier is removed from a data set of 15 values. Which summary changes the most?

(A) the mean

(B) the median

(C) the first quartile

(D) the interquartile range

Letter: **A**

16. Two shops each have a median delivery time of 30 minutes. Shop X's IQR is 4 minutes and Shop Y's is 12 minutes. Which is a correct comparison?

(A) The shops perform the same, because their medians are equal.

(B) Shop Y is more consistent, because its IQR is larger.

(C) Shop X is faster, because its IQR is smaller.

(D) Both typically take about 30 minutes, but Shop X's times are more consistent.

Letter: **D**

Part C — Short Answer & Computation — 6 questions, 8 points each

17. Eighteen **Granite Falls High School** students with part-time jobs recorded the hours they worked last week.

0	8 9
1	0 2 4 5 5 6 8
2	0 0 1 3 4 6
3	0 2
4	8

Key: 1 | 4 means 14 hours

(a) What does the leaf 8 on stem 4 stand for?

One student who worked 48 hours

(b) Median: 19 **hours** Range: 40 **hours**

(c) Describe the distribution — shape, unusual features, center, and spread — in context.

Roughly symmetric with a slight right tail, a gap from 32 to 48 hours, and a possible outlier at 48. The median is 19 hours, and the hours run from 8 to 48 (range 40 hours).

18. Nine members of the Granite Falls book club read these numbers of books this summer, in order: 3, 5, 6, 6, 8, 9, 11, 12, 21.

(a) Find the mean. 9 books

$$\bar{x} = \frac{3+5+6+6+8+9+11+12+21}{9} = \frac{81}{9} = 9 \text{ books}$$

(b) Median: 8 **books** Mode: 6 **books**

(c) Find Q_1 and Q_3. (Leave the median out of both halves.) 5.5 11.5

$$Q_1 = \frac{5+6}{2} = 5.5 \qquad Q_3 = \frac{11+12}{2} = 11.5$$

(d) Five-number summary: 3, 5.5, 8, 11.5, 21 IQR: 6 **books**

19. The Granite Falls cafeteria line took these numbers of minutes to clear on five days: 9, 11, 13, 13, 14. The mean is 12 minutes.

(a) Complete the table.

						Sum
Value x	9	11	13	13	14	60
Deviation $x - \bar{x}$	-3	-1	1	1	2	0
Squared $(x - \bar{x})^2$	9	1	1	1	4	16

(b) Find the variance and the standard deviation. 4 square min 2 min

$$s^2 = \frac{16}{5-1} = 4 \text{ square minutes} \qquad s = \sqrt{4} = 2 \text{ minutes}$$

(c) Interpret s in context.

A typical day's line time is about 2 minutes from the 12-minute mean.

20. The five-number summary of 30 Granite Falls students' daily phone pickups is 18, 42, 55, 66, 110. The three most extreme values are 18, 97, and 110.

(a) Find the IQR and both fences. 24 6 102

$$\text{IQR} = 66 - 42 = 24$$

$$\text{lower fence} = 42 - 1.5(24) = 6 \qquad \text{upper fence} = 66 + 1.5(24) = 102$$

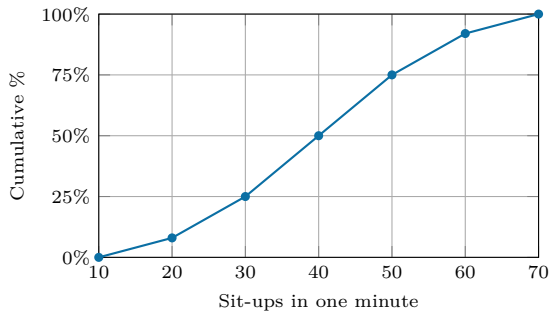
(b) Which of 18, 97, and 110 are outliers? **Only** 110

(c) In a modified boxplot, where does the upper whisker end? **At** 97

(d) The value 18 looks far from the box. Explain whether it is an outlier.

No — 18 is above the lower fence of 6, so it is inside the fences.

21. The graph shows how many sit-ups 100 Granite Falls students did in one minute.



(a) Q_1: 30 Median: 40

Q_3: 50 IQR: 20

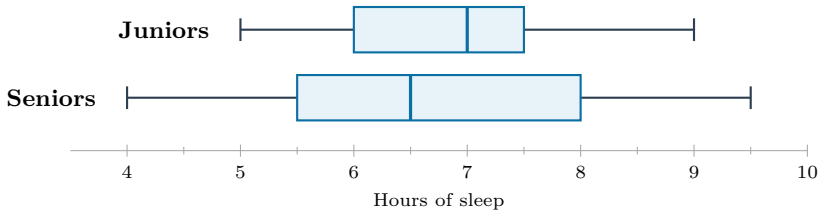
(b) A student did 60 sit-ups. What percentile is that?

92nd

(c) Interpret your answer to (b) in context.

92% of these 100 students did 60 or fewer sit-ups in one minute.

22. Granite Falls surveyed juniors and seniors about hours of sleep on a school night. The parallel boxplots share one scale.



(a) Complete the table.

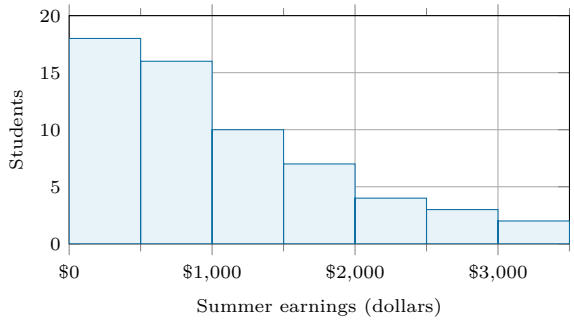
	Median	IQR
Juniors	**7 h**	**1.5 h**
Seniors	**6.5 h**	**2.5 h**

(b) Compare the two groups' sleep, with a comparative word for center *and* for spread.

Juniors' median (7 h) is greater than seniors' (6.5 h), and seniors' sleep is more spread out (IQR 2.5 h vs. 1.5 h), so juniors' sleep is more consistent.

Part D — Extended Response — 3 questions, 10 points each

23. Sixty Granite Falls students reported how much they earned last summer. The histogram and the calculator's 1-Var Stats are below.



1-Var Stats	
$\bar{x}$	1083
Sx	832
n	60
minX	60
Q_1	390
Med	860
Q_3	1620
maxX	3310

(a) Name the shape of the distribution, and justify it from the histogram.

Skewed right: most students earned under $1,000, and the long tail stretches toward the larger earnings, out to about $3,500.

(b) The mean ($1,083) is larger than the median ($860). Explain why, using the shape.

The mean uses every value, so the few students in the long right tail pull it up; the median uses only the middle position, so it stays lower.

(c) Which center and which spread should the school report? Give their values in context and say why.

The median, $860, and the IQR, $1,230 (the middle half earned $390 to $1,620). The data are skewed right, so the tail pulls the mean and SD up.

(d) A student wrote: *"Skewed left — most students earned a little, so the peak is on the left."* Explain what is wrong.

The skew is named by the long tail, not the peak. The pile is on the left, but the tail stretches toward larger earnings, so it is skewed right.

24. Two Granite Falls bowlers each rolled five games. **Maya:** 180, 182, 184, 186, 188 ($\bar{x} = 184$). **Jordan:** 110, 125, 140, 150, 175 ($\bar{x} = 140$, $s \approx 24.7$).

(a) Complete the table for Maya, then find her standard deviation. 3.2 pins

						Sum
Score x	180	182	184	186	188	920
Deviation $x - \bar{x}$	-4	-2	0	2	4	0
Squared $(x - \bar{x})^2$	16	4	0	4	16	40

$$s^2 = \frac{40}{5-1} = 10 \qquad s = \sqrt{10} \approx 3.2 \text{ pins}$$

(b) A friend says, *"Maya's standard deviation should be the bigger one — her scores are way higher."* Use the formula to explain why the friend is wrong.

s is built from the deviations — each score's distance from its own mean. Maya's scores sit within 4 pins of 184; how big the scores are never enters.

(c) Next week every one of Maya's five scores is exactly 10 pins higher. New mean: 194 New standard deviation: 3.2

Explain why the standard deviation came out that way.

Adding 10 to every score adds 10 to the mean, so not one deviation changes.

(d) Which bowler is more consistent? Justify with numbers.

Maya: $s \approx 3.2$ pins against Jordan's 24.7, so her games stay close together.

25. Granite Falls's science department tested 12 batteries of each of two brands. Here are the lifetimes, in hours, from the calculator.

Brand	$\bar{x}$	**Sx**	**minX**	Q_1	**Med**	Q_3	**maxX**
Brand A	40	2.1	36	38.5	40	41.5	44
Brand B	40	7.8	26	34	40	46	52

Both distributions are roughly symmetric.

(a) Check Brand B for outliers with the $1.5 \times$ IQR rule.

IQR $= 46 - 34 = 12$

fences $= 34 - 18 = 16$ and $46 + 18 = 64$

Every value lies between the fences (26 to 52), so Brand B has no outliers.

(b) The department head says, *"Same median, same mean — it doesn't matter which we buy."* Respond.

When the centers match, the spread decides. Brand B's lifetimes are much more spread out, so some B batteries die far sooner than any A battery.

(c) Write a comparison of the two brands — shape, outliers, center, and spread — using comparative words.

Both brands are roughly symmetric with no outliers, and both centers are 40 hours. Brand B is much more spread out (Sx 7.8 h, range 26 h) than Brand A (Sx 2.1 h, range 8 h), so Brand A's lifetimes are more consistent.

(d) A flashlight must run at least 35 hours. Which brand should the department buy? Justify with the table.

Brand A. Its minimum is 36 hours, so every A battery lasted at least 35 hours; Brand B's Q_1 is 34, so at least a quarter of B batteries fall short.